

Automorphism Orbits of Direct Factors in a Restricted Mark Cokernel

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Abstract

An exact restricted mark cokernel C was previously decomposed as $D \oplus K$, where $D \cong \mathbb{Z}/8 \oplus (\mathbb{Z}/2)^2$. We classify the global families generated by this decomposition. The automorphism group of C acts transitively on the D -type direct factors and on the ordered decompositions of type (D, K) and on split embeddings of D . Exact stabilizer and mass formulas give 5846893330432 direct factors and 12857454406351852314558464 ordered decompositions. A subgroup formula counts all D -type subgroups and shows that 2947589144576 are not direct factors, giving the sharp boundary of the orbit statement.

1 Input and scope

The exact primary types, written as exponent partitions, are

$$C_{(2)} : (4, 3, 2, 2, 2, 1^{10}), \quad C_{(3)} : (2, 1),$$
$$D : (3, 1, 1), \quad K_{(2)} : (4, 2, 2, 2, 1^8), \quad K_{(3)} : (2, 1).$$

Thus $C = D \oplus K$. C69 constructed one such decomposition and showed that the fixed D has 2^{41} complements. Here we classify all direct factors and all decompositions of these types.

The scope firewall is

NO_BAD_EULER_OR_ROOT_NUMBER.

We do not claim a canonical decomposition, a full table of marks or Burnside ring, arithmetic or local data, Euler factors, root numbers, automorphy, or a Hilbert–Polya operator.

2 Automorphism and Hom counts

For a finite abelian p -group of exponent type λ , write λ' for the conjugate partition and $m_s(\lambda)$ for the multiplicity of s . Then

$$|\mathrm{Aut}(G_\lambda)| = p^{\sum_r (\lambda'_r)^2} \prod_s \prod_{j=1}^{m_s(\lambda)} (1 - p^{-j}). \quad (1)$$

Indeed, the first power counts endomorphisms. Reduction on the summands of each exact exponent s must be invertible; its relative density is the corresponding product. We also use

$$|\mathrm{Hom}(G_\lambda, G_\mu)| = p^{\sum_{i,j} \min(\lambda_i, \mu_j)}. \quad (2)$$

The conjugate partitions for $C_{(2)}$, D , $K_{(2)}$ are respectively $(15, 5, 2, 1)$, $(3, 1, 1)$, and $(12, 4, 1, 1)$. Exact evaluation gives

$$|\mathrm{Aut}(D)| = 2^7 \cdot 3 = 384, \quad |\mathrm{Aut}(C_{(3)})| = |\mathrm{Aut}(K_{(3)})| = 108,$$

and

$$|\mathrm{Aut}(C)| = 2^{194} 3^{10} 5^2 7^4 11 17 31^2 73 127,$$
$$|\mathrm{Aut}(K)| = 2^{121} 3^9 5^2 7^3 17 31 127.$$

Equation (2) gives

$$\log_2 |\mathrm{Hom}(K, D)| = 3 \cdot 12 + 1 \cdot 4 + 1 \cdot 1 = 41.$$

Although $|\mathrm{Hom}(D, K)|$ happens to have the same value, the setwise stabilizer below specifically uses maps from K to D .

3 Two homogeneous spaces

Any D -type direct factor has some complement. Uniqueness of the cyclic elementary divisors forces every such complement to have type K . Given two ordered decompositions, choose isomorphisms between their D components and between their K components; their direct sum is an automorphism of C . This proves transitivity on ordered decompositions and, after forgetting the complement, on D -type direct factors.

For the fixed ordered pair, the stabilizer is block diagonal:

$$\text{Stab}(D, K) = \text{Aut}(D) \times \text{Aut}(K).$$

The setwise stabilizer of D consists of invertible upper block-triangular maps, hence

$$\text{Stab}(D) = \text{Hom}(K, D) \rtimes (\text{Aut}(D) \times \text{Aut}(K)).$$

If an abstract copy of D is identified with its image, its pointwise stabilizer is $\text{Hom}(K, D) \rtimes \text{Aut}(K)$.

Theorem 1 (Exact orbit masses). *The number of D -type direct factors in C is*

$$N_D = 2^{25} \cdot 7 \cdot 11 \cdot 31 \cdot 73 = 5846893330432.$$

The number of ordered decompositions $C = D' \oplus K'$ of type (D, K) is

$$N_{\text{dec}} = 2^{66} \cdot 7 \cdot 11 \cdot 31 \cdot 73 = 12857454406351852314558464.$$

Proof. Orbit–stabilizer gives

$$N_D = \frac{|\text{Aut}(C)|}{|\text{Aut}(D)| |\text{Aut}(K)| |\text{Hom}(K, D)|}, \quad N_{\text{dec}} = \frac{|\text{Aut}(C)|}{|\text{Aut}(D)| |\text{Aut}(K)|}.$$

Exact integer division yields the displayed values. The independent mass identities

$$N_{\text{dec}} = N_D 2^{41}, \quad |\text{Aut}(C)| = N_D |\text{Stab}(D)| = N_{\text{dec}} |\text{Stab}(D, K)|$$

hold exactly. The first also recovers the C69 complement count for every member of the direct-factor orbit. $\square$

Every direct factor has $|\text{Aut}(D)| = 384$ parameterizations by an abstract D . Thus split embeddings form a third transitive $\text{Aut}(C)$ -space of size

$$N_{\text{emb}} = N_D |\text{Aut}(D)| = 2245207038885888.$$

4 Why “direct factor” is essential

Write the 2-primary diagonal model with a generator g_4 of its $\mathbb{Z}/16$ factor and elementary generators f_1, f_2 from two $\mathbb{Z}/2$ factors. Set

$$H = \langle 2g_4, f_1, f_2 \rangle.$$

Proposition 1 (Isomorphic non-factor). *The subgroup H is isomorphic to D , but it is not a direct factor of C .*

Proof. The three generators have orders 8, 2, 2 and are independent, so H has type $(3, 1, 1)$. Quotienting changes the $\mathbb{Z}/16$ factor to $\mathbb{Z}/2$ and removes two elementary factors. Thus $(C_{(2)}/H)$ has type $(3, 2, 2, 2, 1^9)$, whereas a complement to a D -type direct factor must have type $K_{(2)} = (4, 2, 2, 2, 1^8)$. Hence H cannot be a direct factor. $\square$

The boundary can be counted globally. For p -group types $\mu \subseteq \lambda$, the number of subgroups of type μ is

$$p^{\sum_i \mu'_{i+1}(\lambda'_i - \mu'_i)} \prod_i \begin{bmatrix} \lambda'_i - \mu'_{i+1} \\ \mu'_i - \mu'_{i+1} \end{bmatrix}_p.$$

For $\lambda = C_{(2)}$ and $\mu = D$, this becomes

$$2^{16} \begin{bmatrix} 14 \\ 2 \end{bmatrix}_2 \begin{bmatrix} 2 \\ 1 \end{bmatrix}_2 = 8794482475008.$$

Subtracting the direct-factor orbit leaves exactly 2947589144576 non-direct D -type subgroups. Multiplication by $|\text{Aut}(D)|$ gives 3377081270403072 monomorphisms in total, so the split qualification is also essential for the embedding orbit.

5 Reproducibility and conclusion

The evidence binds the C66 and corrected C69 bytes. The producer evaluates the conjugate-partition formula, while the checker counts endomorphism blocks and their invertible reductions. SymPy checks the large prime factorizations, GAP independently gives $|\text{Aut}(D)| = 384$, clean replay passes, and thirty hostile semantic mutations are rejected. The result is therefore a complete orbit-and-stabilizer classification for direct factors, with no extension to all abstractly isomorphic subgroups.